



Custos User & Role Management Portal

Students: Ruihuan Gao, Jayanth Vennamreddy

Mentors: Lahiru Jayathilake



Georgia Tech
Open Source Program Office

Project Overview

Apache Custos is a security middleware for science gateways and HPC research computing. It provides identity, RBAC(Role - Based Access Control) , credentials, and resource allocation through reusable APIs.

Research sites need one place to manage who can log in, what they can access, and how much compute they can use, instead of building that stack themselves.

Custos had backend RBAC APIs, but the portal did not expose the full admin workflow. This project connected the portal to live Custos APIs so admins can manage users, roles, privileges, and access from the dashboard.

Open Source Outcomes

- PR #524: wired Role Management tab to RBAC APIs
- PR #528: added live User Management role assignment
- both PRs merged into apache:master
- changes were reviewed, revised, tested, and aligned with the existing portal UI

Project Work

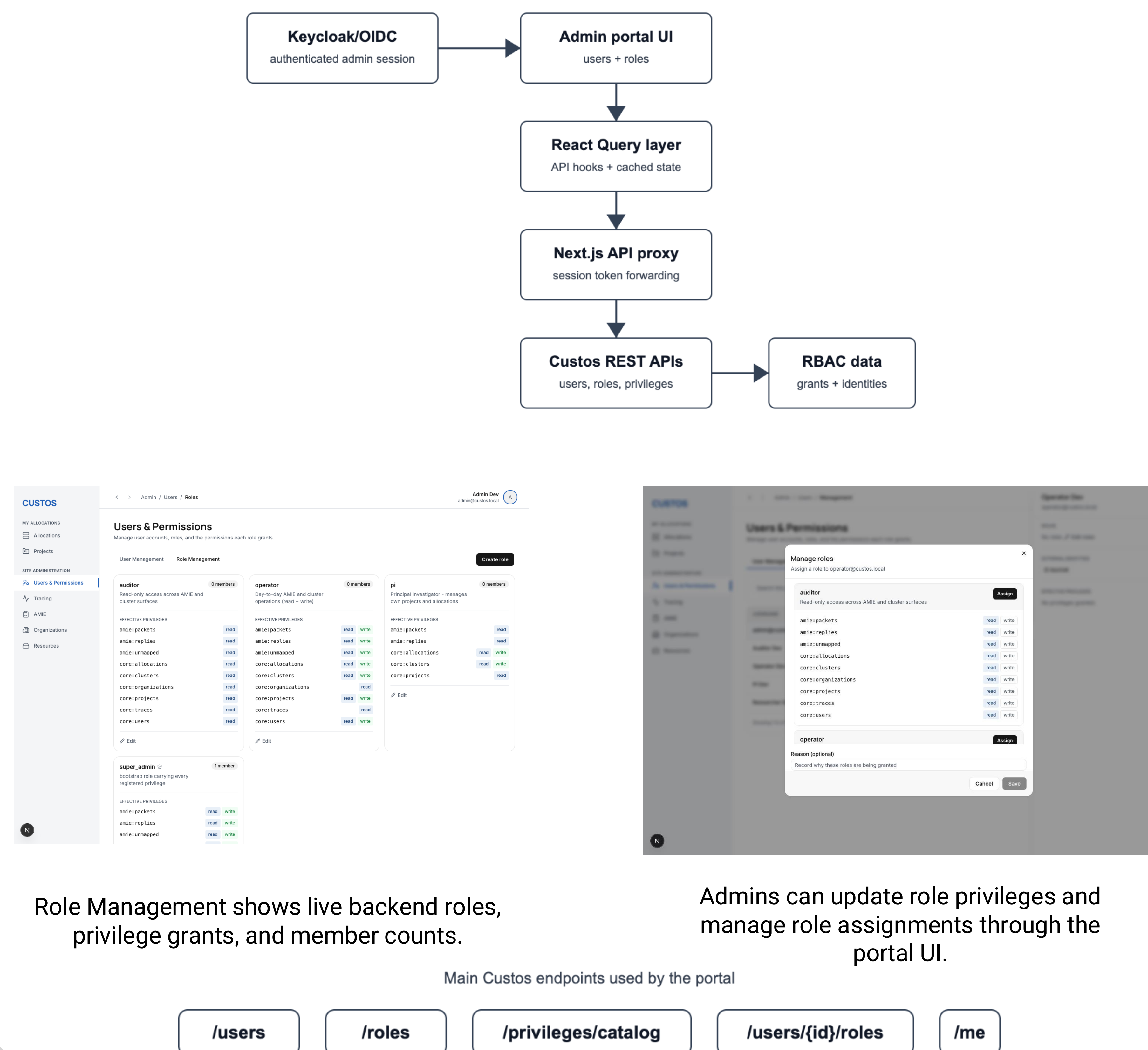
Jayanth: Role Management

- listed live backend roles with privileges and member counts
- added create/update role flows
- wired privilege add/remove through RBAC endpoints
- added assign-role-to-many-users support
- fixed 204 No Content proxy handling found during Keycloak/OIDC (OpenID Connect) testing

Ruihuan: User Management

- loaded users, identities, roles, and privileges from live APIs
- added multi-role assign/unassign from Manage Roles
- added rollback handling for failed role updates
- added user-management tests and proxy coverage

Highlights and Accomplishments



Testing and Validation

Validation included typecheck, lint, unit tests, Playwright e2e tests, backend tests, and local Keycloak/OIDC testing with MSW(Mock Service Worker) disabled. The live auth setup helped catch an API proxy issue with empty backend responses, which was fixed and covered with a regression test.

Sources

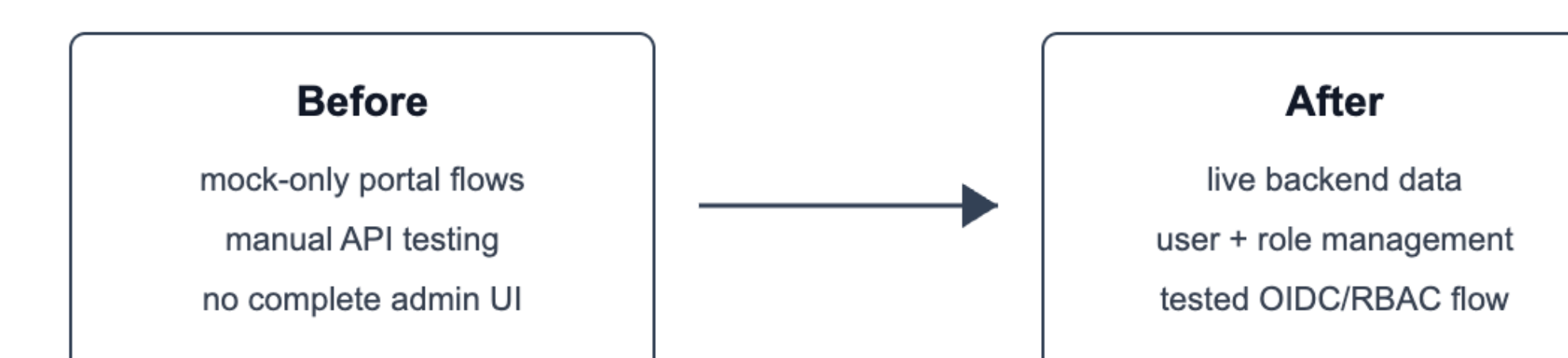
Learn more at <https://airavata.apache.org/custos/>

[PR #524: Role Management tab to RBAC APIs](https://github.com/apache/airavata-custos/pull/524) (<https://github.com/apache/airavata-custos/pull/524>)

[PR #528: User Management role assignment](https://github.com/apache/airavata-custos/pull/528) (<https://github.com/apache/airavata-custos/pull/528>)

Goals

- connect User Management and Role Management to live Custos APIs
- support admin workflows for roles, privileges, identities, and user role assignments
- keep the existing Custos RBAC model as the source of truth
- add tests for the main UI flows and API proxy behavior



Future Work / Next Steps

- add a batch role-assignment API so multi-role updates can be handled in one backend operation
- reduce partial-update risk from sequential role assignment calls
- expand the end-user My Account / My Access views
- continue staging validation with Keycloak/OIDC and RBAC permission checks

